HW 15

Due Thursday, July 23 at 11:59pm.

AI use

Refer to the syllabus for our AI use policy. You will be asked to include any AI prompts used in this assignment at the end.

Starting a new Quarto file

For this assignment you will begin a new Quarto file. To do this, go to RStudio and click File > New File > Quarto Document. A pop-up will appear asking for a title, author, and format. You may leave these as the default options for now. Click OK and a new file will open. Update the YAML with your title, name, and date.

In this assignment, you will work with a **simulated environmental dataset** containing four variables:

- `bird_presence`: whether a rare bird species was detected at a survey site (0/1)
- `air_quality`: numeric index (higher = worse air quality)
- `trees`: number of trees per hectare at the site
- `wetland`: whether the site is a wetland (1 = yes, 0 = no)

Computational setup

```r
library(tidymodels)
library(kableExtra)
```

Simulating the dataset

Run the following code to simulate the dataset.

```r
set.seed(9745)

n <- 200

air_quality <- rnorm(n, mean = 50, sd = 10)      # continuous predictor
trees <- rpois(n, lambda = 30)                   # count predictor
wetland <- rbinom(n, size = 1, prob = 0.4)       # binary predictor

# true model:
# logit(p) = -7 + 0.05*air_quality + 0.04*trees + 1.0*wetland
log_odds <- -7 + 0.05 * air_quality + 0.04 * trees + 1 * wetland
prob <- exp(log_odds) / (1 + exp(log_odds))

bird_presence <- rbinom(n, size = 1, prob = prob)
```

```r
env_data <- data.frame(
  bird_presence,
  air_quality,
  trees,
  wetland
)

head(env_data)
```

```
  bird_presence air_quality trees wetland
1             0    37.55806    25       0
2             0    45.48046    24       1
3             1    44.22666    40       0
4             0    42.80878    26       0
5             0    45.46971    38       0
6             0    33.00507    37       0
```

Exercise 1

Fit a logistic regression model predicting bird presence from air quality, trees, and whether the side was a wetland. Display the coefficient estimates table using `kable()`, rounding to 3 digits.

Exercise 2

Interpret the estimated coefficient for `air_quality` in terms of **odds**. (You can state in terms of a multiplicative change or a percent difference in odds.) (1–2 sentences)

Exercise 3

Interpret the estimated coefficient for `wetland` in terms of **odds**. (You can state in terms of a multiplicative change or a percent difference in odds.) (1–2 sentences)

Exercise 4

Predict the **probability** of bird presence for a site with:

- `air_quality` = 60
- `trees` = 40
- `wetland` = 1 (wetland site)

Use `predict(..., type = "response")`.

AI Attestation

Was AI used on this assignment? If so, describe your prompts below. If not, simply state "AI was not used on this assignment."

Submission

As you've seen previously, we can **Render** into an .html file that can be opened by any web browser. To export it as a .pdf, open the file in your web browser and then print to or save as a .pdf document. Contact me in Ed Discussion if you need help!

You will submit the PDF documents for labs and homework to Gradescope as part of your final submission.

To submit your assignment:

- Access Gradescope through the menu on the BIOS 600 Canvas site.
- Click on the assignment, and you'll be prompted to submit it.
- Mark the pages associated with each exercise. All of the pages of your lab should be associated with at least one question (i.e., should be "checked").
- Select the first page of your .PDF submission to be associated with the "Formatting" section.

Grading

Component	Points
Ex 1	2
Ex 2	2
Ex 3	2
Ex 4	2
AI Attestation	1
Formatting	3

The "Formatting" grade is to assess the document format. This includes having a neatly organized document (no excessive output, warnings/messages when loading packages and/or data) with readable code and your name and the date updated in the YAML.